

Simulation and Modeling Lab -7B2

Question 1: A university claims that the **average time students spend completing a laboratory experiment is 45 minutes**. A random sample of 8 students recorded the following completion times:

- **Data:** [42, 47, 44, 49, 46, 43, 48, 45]
- **Tasks:**
 - Calculate the **t-statistic and p-value**.
 - Determine whether the population mean is significantly different from the claimed mean at a **0.05 significance level**.
 - **Interpret and visualize** the result.

Question 2: An experiment compares the processing time of two algorithms. The measured processing times (in milliseconds) are:

- **Algorithm A:** [42, 45, 39, 44, 41, 43, 40]
- **Algorithm B:** [48, 51, 46, 49, 50, 47, 52]

Assume **equal population variances** and use $\alpha = 0.05$.

Determine whether there is a significant difference between the average processing times of the two algorithms.

Tasks:

1. State **H_0 and H_1** .
2. Calculate the **sample means and sample variances (ddof = 1)**.
3. Calculate the **mean difference** between Algorithm A and Algorithm B.
4. Determine the **pooled variance and standard error**.
5. Calculate the **t-statistic and degrees of freedom**.
6. Find the **two-tailed p-value**.
7. Construct the **95% confidence interval** for the mean difference.
8. **Decision & Interpretation:** State whether the evidence suggests that the two algorithms have different mean processing times.

